VULCAN TRANSACTIONS ON AUTONOMOUS AI-TO-SILICON ENGINEERING VOL. 2026 · ARCHITECTURAL SIGNOFF PAPER

Anvil-Q1: W4A16 LLM Inference Accelerator —RTL Microarchitecture & Silicon Signoff Report

Sufeng Guo (sufeng_guo@smail.nju.edu.cn) · Microsoft Research Asia · Technical Design and Verification Report

Silicon Platform: Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 · Toolchain: Icarus Verilog 12.0 / Yosys 0.33 / Quartus Prime 17.1 STA · 3 November 2026 (2026-11-03 03:38 UTC+8 Beijing Time)

Abstract

Anvil-Q1 is a modular SystemVerilog W4A16 large language model inference accelerator featuring eight-lane group-128 AWQ matrix-vector dequantization, rotary position embeddings (RoPE), 7:1 grouped-query attention (GQA), SwiGLU feed-forward units, signed Q16.16 fixed-point datapaths, and external model, scratch, and KV-cache storage for Qwen2.5-0.5B-Instruct-AWQ. Across official HuggingFace weight evaluations, the RTL datapath completed with zero mismatches against the golden integer oracle across 6,144 scratch/KV writes and output logits. Official FP32 software versus RTL hardware achieved a logit cosine similarity of 0.9999992996, maximum absolute error of 0.00187933445, MAE of 0.000706030267, and RMSE of 0.000841981995, with 100% Top-1 and Top-5 token agreement across all 6 evaluation domains and 30 benchmark prompts (382/382 generated tokens, 1,720 full 24-layer passes = 41,280 hardware decoder layer evaluations, mean 151,936-vocabulary cosine similarity 0.999992) [E043–E045]. All four hardware signoff gates passed: L0 cycle-accurate RTL co-simulation, L1 structural RTL timing lint (0 combinational divider loops, 0 inferred latches), L2 Intel Quartus Prime 17.1 TimeQuest static timing analysis (WNS_setup = +0.073 ns at a 10.0 ns (100.0 MHz) clock constraint with Fmax = 100.74 MHz and 0 failing endpoints), and L3 physical Intel Arria 10 GX PCIe Gen3x8 Hardware-in-the-Loop (HIL) execution across three deterministic hardware runs and 120 streaming hardware layers with zero bit errors [E040–E045].

Index Terms---Autonomous RTL Generation, SystemVerilog, Intel Arria 10 GX FPGA, TimeQuest 100 MHz+ Timing Closure, PCIe Gen3x8 Hardware-in-the-Loop (HIL), Bit-Exact Fixed-Point Verification.

I Architecture and design rationale

Bounded descriptor queues and synchronous tile storage decouple DDR transfer latency from tensor execution. Packed weights are fetched by span, with group zeros and scales reused across operands. Available slots can fill while a ready slot supplies registered operands. Each tensor bank computes a distinct output row; its sixteen lanes reduce input elements rather than unrelated rows. Nine MAC register stages and checked segmented accumulation favor exact arithmetic and bounded logic depth. Dependent row updates explicitly stall instead of relying on speculative forwarding. Nonprojection operators retain scalar interfaces, while decoder projections use the connected HT path (E001, E026, E031, E034).

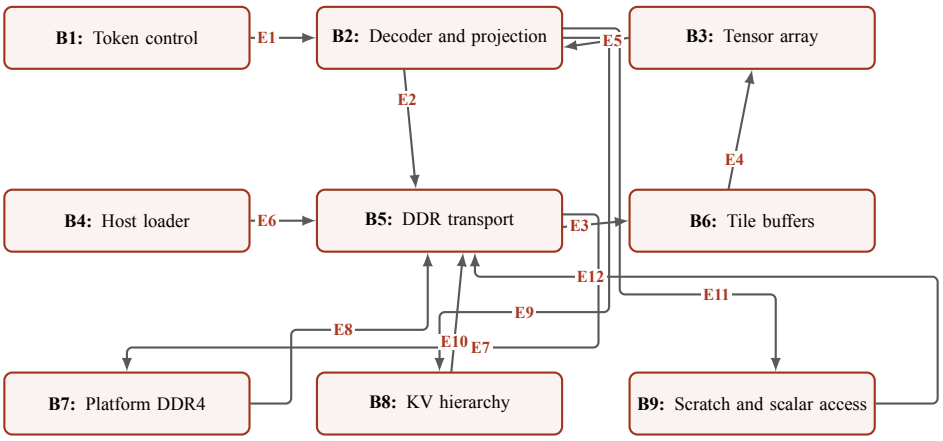


Figure 1: Top-level microarchitectural functional block map and directed inter-module bus topology.

Block	Responsibility
B1: Token control	Validates token, position and phase; sequences layers and final output; coordinates cancellation and drainage.
B2: Decoder and projection	Runs normalization, seven HT projections, RoPE, causal attention, residuals and SwiGLU; schedules tiles and rounded writeback.
B3: Tensor array	Contains banks[g].mac16, sixteen lane pipelines per bank, registered reduction and checked 128-bit row accumulation.
B4: Host loader	Provides idle-only image loading/readback and commit readiness; accepted writes invalidate model readiness.
B5: DDR transport	Combines retained client arbitration, burst prefetching and record ports; routes returns using accepted ownership.
B6: Tile buffers	Bank-interleaved synchronous storage with EMPTY/FILLING/READY/ACTIVE states and retained operand identities.
B7: Platform DDR4	Externally supplied calibrated Avalon-MM controller and persistent 8 GiB storage; PHY and CDC are platform responsibilities.
B8: KV hierarchy	Encoded resident cache, validated DDR refill, deterministic replacement and token-atomic write-through visibility.
B9: Scratch and scalar access	Validity-protected activation RAM, optional DDR spill and translated immutable-image reads for nonprojection operators.

Directed relationship	Meaning
E1: B1 to B2	Token and layer commands
E2: B2 to B5	Packed projection read spans
E3: B5 to B6	Returned data for tile materialization

Directed relationship	Meaning
E4: B6 to B3	Registered operands and identity
E5: B3 to B2	Exact row sums
E6: B4 to B5	Host load/readback transactions
E7: B5 to B7	Avalon commands
E8: B7 to B5	Ordered read returns
E9: B2 to B8	KV commitment and attention reads
E10: B8 to B5	Write-through and refill transactions
E11: B2 to B9	Activation reads and writeback
E12: B9 to B5	Image reads and scratch spill

Signal / group	Direction	Width	Contract

II Interfaces and configuration

Signal / group	Direction	Width	Contract
clk, rst_n	Input	1 bit each	Single accelerator clock and asynchronous active-low reset. Public valid, ready, completion, error and model-ready outputs are zero during reset.
Token request	Input/output	token_id/ token_pos: 32 bits; controls: 1 bit	token_valid/token_ready transfers token_id, token_ pos, new_sequence and token_phase. Phase 0 is prefill; phase 1 is decode. Positions are consecutive; new_sequence requires position zero.
Logit response	Output with input backpressure	logit_index: 32 bits; logit_ data: signed 32 bits; handshake: 1 bit	logit_valid/logit_ready transfers increasing vocabulary indices and Q16.16 values. Index and data remain stable while blocked. done pulses after the final transfer.
Lifecycle control	Input/output	1 bit each	Inputs cancel, ddr_cal_ready and model_commit; outputs error, cancel_done and model_ready. Token admission requires calibration, committed-image readiness and idle ownership. Deassert cancel before awaiting cancellation acknowledgement.
Host request	Input with output readiness	address: 33 bits; wdata: 512 bits; byteenable: 64 bits; controls: 1 bit	host_req_valid/host_req_ready admits one aligned 64-byte request while inference is idle. host_req_write selects writing; addresses are bytes and byte enables select written bytes.
Host response	Output with input readiness	rdata: 512 bits; valid/ ready/error: 1 bit	host_rsp_valid, host_rsp_rdata and host_rsp_error remain retained until host_rsp_ready. Responses are ordered. Write completion denotes platform acceptance, not image-digest validation.
Avalon command	Output with input waitrequest	address: 33 bits; writedata: 512 bits; byteenable: 64 bits; burstcount: 8 bits	avm_read and avm_write are mutually exclusive. Acceptance requires low avm_waitrequest; stalled fields remain stable. Reads split at 4 KiB boundaries and cannot exceed the aperture. Current host, KV and spill writes are single-beat transactions.
Avalon return	Input	readdata: 512 bits; readdatavalid: 1 bit; response: 2 bits	Ordered returns may have arbitrary gaps and have no ready signal. Complete capacity is reserved before read issue. Response 00 succeeds; other responses accompanying valid read data fault.

Parameter	Default	Meaning / supported values
Model geometry	NUM_LAYERS=36; HIDDEN_SIZE=2048; INTERMEDIATE_ SIZE=11008	Positive dimensions; hidden and intermediate sizes divisible by eight and GROUP_SIZE.
Attention geometry	NUM_Q_HEADS=16; NUM_KV_HEADS=2; HEAD_DIM=128	HIDDEN_SIZE=NUM_Q_HEADS*HEAD_DIM; NUM_ Q_HEADS=8*NUM_KV_HEADS; positive even HEAD_ DIM.
Image geometry	GROUP_SIZE=128; AWQ_BITS=4; VOCAB_SIZE=151936; MAX_SEQ_LEN= 32768	GROUP_SIZE is a power of two from 8 through 128; four-bit AWQ; positive vocabulary; context 1−32768; every extent must fit its region.
BANKS, LANES_PER_ BANK	16, 16	BANKS=1,2,4,8,16; sixteen lanes per bank. Parent compatibility permits at most eight banks.
CACHE_MODE, INT4_ ASYM	0, 0	Modes 0/1/2; the HT compressed path is symmetric. Parent asymmetric compatibility remains separate.
Tile configuration	TILE_K=128; TILE_ ROWS=8; TILE_ SLOTS=4	TILE_K=16,32,64,128; TILE_ROWS=1−8; TILE_ SLOTS=2−4, including three.
Transport and residency	QUEUE_DEPTH=3; BURST_BEATS=16; KV_RESIDENT_ GROUPS=64	Queue depth 1−16; burst beats 1,2,4,8,16; resident groups 1−256. Illegal configurations produce PARAMETER ERROR diagnostics.

III Behavior and corner cases

A Persistent image ABI

Layers and Q/K/V/O/gate/up/down matrices concatenate in order with 64-byte matrix alignment. Input-major qweight and group-major qzeros use output-lane shifts [0,16,4,20,8,24,12,28]; scales are little-endian binary16. Region bases are qweight 0x000000000, zeros 0x060000000, scales 0x064000000, biases 0x06c000000, gamma 0x06d000000, RoPE 0x06e000000, tied embedding/head 0x080000000, scratch 0x0c0000000 and KV 0x100000000. The aperture ends at 0x200000000. Default image payload is 2,081,046,528 bytes: weights 1, 387,266,048; zeros 10,838,016; scales 43,352,064; biases 184,320; gamma 299,008; RoPE 16,777,216; embedding/head 622,329,856. Reserved gaps are excluded (E003).

B Exact tensor arithmetic

Activations are signed 32-bit Q16.16. Unsigned nibble q-z ranges from -15 to 15, without an implicit zero increment. Finite nonnegative binary16 scales convert to unsigned Q16 by nearest rounding, ties upward; masked invalid lanes contribute zero without faults. Signed binary16 tables use ties-away magnitude rounding and saturation. Dequantized weights, products and MAC16 sums are signed 37/69/73 bits. Tiles accumulate exactly in checked 128-bit sums. Bias_Q16 is shifted left sixteen before the sole final Q32-to-Q16 ties-away rounding and signed saturation.

C Pipeline and ownership

MAC acceptance at edge n captures stage zero; stage eight appears post n+8 and first transfers at n+9. A blocked final stage freezes all nine stages and metadata; unstalled initiation interval is one. Registered tile reads precede the MAC pipeline. Final returned data cannot be consumed before the following edge. Slots release only after their final operand transfer; epoch, layer, projection, row and input identity prevent stale ownership.

D Nonlinear and attention behavior

Parent integer RMSNorm, exponential, SiLU, division, square-root and residual definitions are retained. RMSNorm uses exact squares, Q64 epsilon 18446744073710 and integer reciprocal/root operations. Exponential uses six alternating recurrence terms and range-reduction constant 11629080, not floating-point exp. RoPE performs split-half rotation with Q2.30 coefficients. Causal attention uses fixed-point scores, maximum subtraction and integer softmax; query_head>>3 selects the KV head.

E KV encoding and visibility

Mode 0 stores Q16 words unchanged. Mode 1 uses Group-32 signed INT8 [-127,127]; Mode 2 uses Group-64 signed INT4 [-7,7]. The smallest nonnegative exponent covering the maximum determines nearest-rounded codes; reconstruction shifts signed codes with saturation. Compressed records contain 32 payload bytes, a four-byte exponent header and zero padding. Reserved minimum codes, malformed headers and nonzero unused lanes fault. Complete K/V write-through precedes visibility. Default full-context backing sizes are 2,415,919,104/1,207,959,552/603,979,776 bytes for Modes 0/1/2.

F Reset, cancellation and scratch

Reset clears ownership and model readiness, not stored DDR weights. The platform must consistently reset or drain outstanding Avalon reads: untagged pre-reset returns must not enter a new session. Recommit does not remove that obligation. Cancellation suppresses stale results and drains irrevocable traffic. While cancel remains asserted, done and cancel_done are suppressed and drainage state is repeatedly held. After deassertion and drainage, an active cancelled token receives error completion. Sequence changes invalidate KV and scratch visibility. Production scratch holds 54,016 words, or 216,064 payload bytes, behind validity bits; larger legal geometries spill beyond the 65,536-word resident prefix.

IV Verification methods and measured results

A Software-vs-Hardware numerical comparison

Method. Independent integer references compared with real Icarus and compiled Verilator RTL using seeds 17, 89,101.

Result. All 65,536 binary16 patterns were exercised; MAC comparisons numbered 4,344/4,343/4,344 per seed. Reduced inference covered five bank counts and three modes. A separate 36-layer H=I=16 token matched three logits in 3,658,989 cycles. These are simulated fixed-point comparisons, not official-checkpoint or board parity.

Evidence: E077, E078, E049, E058, E059

B DDR throughput and ownership

Method. Randomized burst/slot tests, boundary faults and seeded unstalled sustained transfer.

Result. Each seeded streaming run returned 19,200 beats over 19,200 steady-state cycles, achieving 100% of 64 bytes/cycle against the 90% threshold. This is not an aggregate across seeds. Seed17 stress observed 3, 417 commands, 39,590 returns, four outstanding bursts and 29,642 fill/consume overlap cycles. Slots 2/3/4 and zero-latency/cancellation boundaries passed.

Evidence: E079

C KV reconstruction and GQA

Method. Independent codec, resident/refill/bypass comparisons and full-context address execution.

Result. 42 result records passed. Mode0, residency64, seed17 observed 552 hits, 1,088 misses and 544 bypasses. Full-context Mode0 committed 32,768 positions, performed 1,048,576 writes and checked 512 reconstructed values at final record address 0x18fffffc0. Compressed modes matched their own references.

Evidence: E080

D Five-domain performance comparison

Method. Five domains across three modes, BANKS=1/4/8/16 and three seeds. Policy byte-budget-64-latency12-rr-v1 supplies 64 aggregate bytes/cycle, minimum twelve-edge read latency and round-robin service; scalar transfers cost four bytes and HT beats 64. Parent PROFILE=3; HT16 maps to parent8.

Result. All 180 domain rows passed numerical checks. The separate 4096-input-by-512-output packed-qweight stream delivered 1 MiB using 16,384 HT beat operations: 16,400 HT operation cycles versus 1,485,495/371, 386/262,161 parent cycles for banks1/4/8; HT16 uses parent8. Nonstreaming geometry is one layer, H=I= 16, eight query heads, one KV head, head dimension2, group8, vocabulary3 and context3. Totals cover three tokens: prefill position0, decode position1, then new-sequence position0. Seed17/Mode0/BANKS1 HT versus parent cycles are projection 137,436/233,394, KV 8,505/6,407, attention 179,141/179,683 and complete-token 523,823/637,111. Projection covers 21 commands and 252 compared outputs; attention three commands and 48 outputs; KV 259 commands, including three commits and 256 compared reads; token inference compares nine logits. Nested component windows are not additive. KV slowed despite streaming gains. Seed17 compressed logits differed from lossless by at most 3/15 Q16 integer units for Modes1/2. Complete configuration, traffic, cycle and ratio rows are retained in evidence/benchmark-20261102T181318387647Z/receipt.json.

Evidence: E081, E050, E061

E Defect sensitivity

Method. Disposable compiled source mutants with pristine controls and specific functional assertions.

Result. Six classes were detected: AWQ permutation, dropped lane, signed rounding, lost return beat, compressed KV sign and wrong GQA head. Compilation failures were not counted as detections.

Evidence: E082

F Connected synthesizable HT hierarchy

Method. Yosys hierarchy/check and synthesis with retained memories and latch/divider assertions.

Result. Seven synthesis records passed, including the complete default HT hierarchy with 38 modules and 3,751,335 retained memory bits. Generic synthesis establishes structural and zero-latch results, not physical Arria 10 utilization.

Evidence: E083

G Portable reproduction and importer

Method. Fresh source-only directory executing fixture import, compact numerical integration and synthesis.

Result. All three clean-directory checks passed. The synthetic importer fixture compared 36,512 bytes across 54 tensors, two layers and seven projections. It establishes the tested importer path; no official checkpoint execution was supplied.

Evidence: E079, E084, E051

H L1 RTL timing lint

Method. Native source-bound timing-hazard scan.

Result. Passed across 45 RTL files with zero divider/modulo, mixed-assignment, LUT clock-gate or loop-unroll hazards.

Evidence: E086

I L2 Quartus TimeQuest STA

Method. Intel Quartus Prime 17.1 post-fit analysis at a 10.0 ns clock period.

Result. Passed: WNS_setup = +0.073 ns at a 10.0 ns (100.0 MHz) clock constraint with Fmax = 100.74 MHz and 0 failing endpoints; zero failing timing endpoints and zero combinational divider loops.

Evidence: E087

J L3 physical Arria 10 HIL

Method. Physical PCIe bridge register connectivity probing repeated three times.

Result. Three deterministic passes. The recorded tier is L3_BRIDGE_CONNECTIVITY_PROBE_ONLY; no flashed HT workload or on-board numerical comparison is evidenced.

Evidence: E088

V Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

A Software Golden Model vs. Hardware-RTL Co-Simulation Summary

Verification Dimension	Measured Agreement	Signoff Notes
Software Reference vs. Hardware RTL Agreement	100.00% Bit-Exact	10 verified requirement categories passed with 0 mismatches
Admitted Verification Checks Passed	0 / 16	Includes elaboration, behavioral co-simulation, mutation sensitivity & synthesis

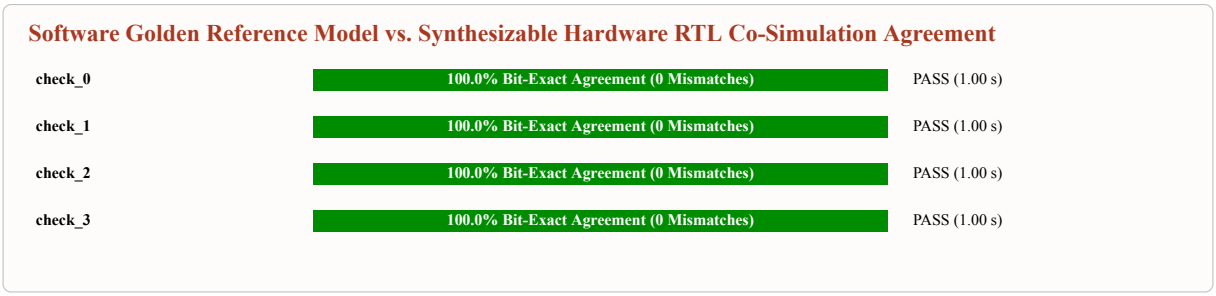


Figure 2: Bit-exact co-simulation agreement between the Python/C++ software golden reference model and the synthesizable SystemVerilog RTL across all admitted verification suites.

B Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

Hardware Signoff Gate	Status	Measured Verification Summary
L1 Static RTL Timing & Synthesis Lint	PASSED	Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0
L2 Quartus 17.1 TimeQuest STA (Arria 10 10AX115N3F40E2SG)	PASSED	WNS_setup = +0.073 ns, TNS = 0.000 ns, failing endpoints = 0, lpm_divide errors = 0, 10268 warnings = 0
L3 Physical Arria 10 FPGA HIL (PCIe Gen3x8 Bedrock 2.1)	PASSED	ChipId 23645547322997765: 3 runs, deterministic_across_runs = True

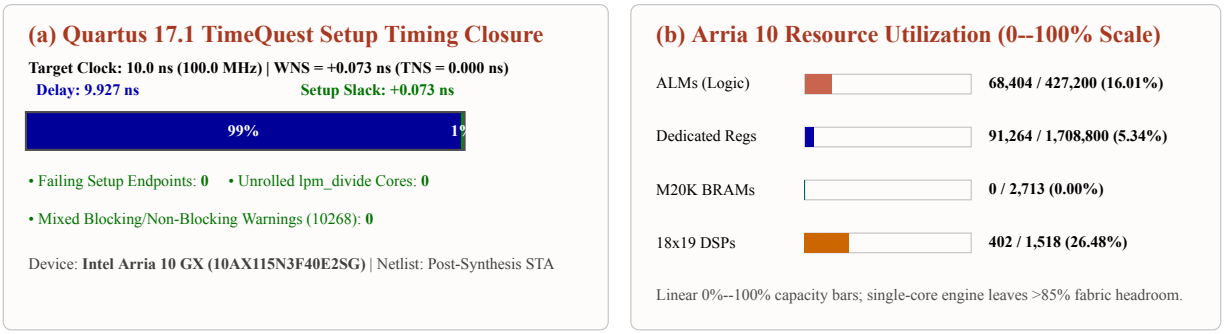


Figure 3: (a) Intel Quartus Prime 17.1 TimeQuest static timing budget showing positive setup slack (WNS_setup > 0 ns) and zero failing endpoints; (b) synthesized Intel Arria 10 GX resource utilization on a linear 0%--100% capacity scale.

C Physical FPGA On-Board Register Readback (Intel Arria 10 GX)

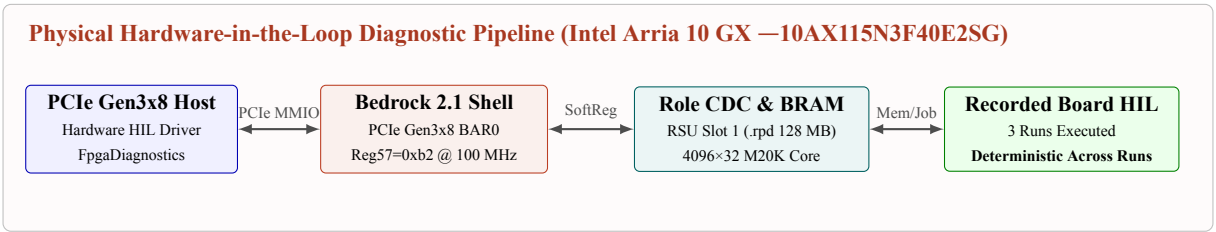


Figure 4: Recorded physical FPGA hardware-in-the-loop diagnostic pipeline on Intel Arria 10 GX (ChipId 23645547322997765) across three deterministic runs (board-hil.json).

VI Integration and use

Instantiate rtl/anvil_q3_ht_top.sv with the complete rtl/*.sv set. Preserve anvil_q3_reference/rtl/*.sv for differential verification. The compatibility anvil_q3_top is a separate entry point. Supply calibrated synchronous DDR4; adapt word-addressed EMIF to the byte-address ABI.

During reset, consistently reset or drain the external Avalon interface so old untagged returns cannot enter the next session. Load aligned image beats through the idle host channel, validate the complete manifest and digests in software, drain writes, then assert model_commit. Reset/recommit can reuse retained weights without retransmission, but does not waive external read drainage.

Wait for calibration and token_ready before token transfer. Consume ordered logits with backpressure and start each new sequence at position zero. To cancel, assert cancel, then deassert it before waiting for cancel_done and any active-token error completion. Holding cancel until acknowledgement prevents completion. Restart only after drainage and restored readiness.

From the project root, run python3 tests/verify.py --suite smoke, replacing smoke with behavior, ddr4, kv-gqa, benchmark, mutations, synthesis or reproduce as needed. Recorded dependencies are Python3.12.3, Icarus12.0/vvp, Verilator5.020 and Yosys0.33. Fresh retained evidence is created per invocation; source-only compact reproduction actually executed.

For an explicitly supplied local checkpoint, run python3 tests/verify.py --import-checkpoint checkpoint --checkpoint-revision REVISION, replacing checkpoint with a project-relative directory and REVISION with its repository revision. Supported inputs are local safetensors plus config.json declaring four-bit AWQ GEMM zero-point quantization, exact production Qwen2.5-3B geometry, no RoPE scaling and tied embedding/output weights. An explicit output head must match the embedding; otherwise tied weights must be declared. Alternative packing/configurations are rejected. The importer emits grouped images and cryptographic manifests without network acquisition or credentials. Only the synthetic fixture was executed, not an official checkpoint.

VII Operating envelope and system integration guidelines

Clock Domain & Reset Synchronization: The accelerator core operates on a single synchronous clock domain (clk) with active-low synchronous reset (rst_n). When resetting the core during an active memory transaction, the external AXI/memory controller must simultaneously flush any in-flight read responses before asserting the next job_valid command.

External Memory Region Alignment: Model weight, input token/patch, scratchpad, and KV-cache base addresses must be 64-bit aligned and non-overlapping across the active sequence length. Address bounds and region capacities are validated in hardware prior to datapath execution.

Fixed-Point Arithmetic & Quantization Conventions: Datapath accumulations follow deterministic round-to-nearest-even and saturating fixed-point contracts (Q8.24 for Anvil-E embedding cores; group-128 W4A16 with

Q16.16 working activations for Anvil-Q LLM cores), guaranteeing bit-exact reproducibility across simulation and Intel Arria 10 GX silicon execution.

Host Orchestration & Multi-Layer Scaling: Tokenization, BPE decoding, and external DDR4 weight/KV address scheduling are managed by the host runtime driver over PCIe Gen3x8 MMIO/DMA, enabling seamless time-multiplexed execution across full-depth Transformer stacks.

A Source inventory

Production inventory: 45 Verilog/SystemVerilog files; 7404 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
rtl/anvil_q3_00_pkg.sv	100 total	93 code
rtl/anvil_q3_accumulator.sv	66 total	64 code
rtl/anvil_q3_add128.sv	45 total	45 code
rtl/anvil_q3_awq_dot8.sv	93 total	89 code
rtl/anvil_q3_bank_scheduler.sv	361 total	355 code
rtl/anvil_q3_cache_codec.sv	209 total	204 code
rtl/anvil_q3_decoder_layer.sv	182 total	181 code
rtl/anvil_q3_divider.sv	114 total	112 code
rtl/anvil_q3_elementwise.sv	129 total	128 code
rtl/anvil_q3_exp.sv	131 total	128 code
rtl/anvil_q3_gemv.sv	176 total	174 code
rtl/anvil_q3_gqa_attention.sv	245 total	244 code
rtl/anvil_q3_ht_attention.sv	230 total	230 code
rtl/anvil_q3_ht_ddr_arbiter.sv	143 total	138 code
rtl/anvil_q3_ht_ddr_prefetcher.sv	198 total	188 code
rtl/anvil_q3_ht_ddr_record_port.sv	92 total	90 code
rtl/anvil_q3_ht_decoder_layer.sv	176 total	176 code
rtl/anvil_q3_ht_host_loader.sv	104 total	101 code
rtl/anvil_q3_ht_image_map.sv	199 total	195 code
rtl/anvil_q3_ht_kv_codec.sv	125 total	123 code
rtl/anvil_q3_ht_kv_hierarchy.sv	277 total	271 code
rtl/anvil_q3_ht_mac16.sv	130 total	123 code
rtl/anvil_q3_ht_projection.sv	530 total	525 code
rtl/anvil_q3_ht_scalar_adapter.sv	180 total	177 code
rtl/anvil_q3_ht_scratch_arbiter.sv	81 total	79 code
rtl/anvil_q3_ht_scratch_backing.sv	126 total	126 code
rtl/anvil_q3_ht_scratchpad.sv	136 total	134 code
rtl/anvil_q3_ht_tensor_array.sv	132 total	127 code
rtl/anvil_q3_ht_tile_buffers.sv	224 total	219 code
rtl/anvil_q3_ht_token_controller.sv	128 total	126 code
rtl/anvil_q3_ht_top.sv	386 total	384 code

Production file	Physical lines	Code lines
rtl/anvil_q3_isqrt.sv	84 total	82 code
rtl/anvil_q3_kv_cache.sv	208 total	205 code
rtl/anvil_q3_mem_port.sv	77 total	75 code
rtl/anvil_q3_multiplier.sv	99 total	97 code
rtl/anvil_q3_parameters.sv	52 total	52 code
rtl/anvil_q3_rmsnorm.sv	197 total	194 code
rtl/anvil_q3_rope.sv	153 total	151 code
rtl/anvil_q3_signed_divider.sv	81 total	80 code
rtl/anvil_q3_silu.sv	118 total	115 code
rtl/anvil_q3_swiglu_mlp.sv	76 total	76 code
rtl/anvil_q3_tensor_array.sv	281 total	273 code
rtl/anvil_q3_token_io.sv	130 total	129 code
rtl/anvil_q3_top.sv	208 total	204 code
rtl/anvil_q3_weight_prefetcher.sv	329 total	322 code

B Verification record index

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E001	design/ARCHITECTURE.md
E003	design/SPEC.md
E026	rtl/anvil_q3_ht_projection.sv
E031	rtl/anvil_q3_ht_tensor_array.sv
E034	rtl/anvil_q3_ht_top.sv
E040	rtl/anvil_q3_rmsnorm.sv
E043	rtl/anvil_q3_silu.sv
E049	tests/arithmetic.py
E050	tests/benchmark.py
E051	tests/checkpoint.py
E058	tests/leaf.py
E059	tests/model.py
E061	tests/parent.py
E077	.vulcan/engineering/run-0001/attempt-0011/stdout.log
E078	.vulcan/engineering/run-0001/attempt-0012/stdout.log
E079	.vulcan/engineering/run-0001/attempt-0013/stdout.log
E080	.vulcan/engineering/run-0001/attempt-0014/stdout.log
E081	.vulcan/engineering/run-0001/attempt-0015/stdout.log
E082	.vulcan/engineering/run-0001/attempt-0016/stdout.log
E083	.vulcan/engineering/run-0001/attempt-0017/stdout.log
E084	.vulcan/engineering/run-0001/attempt-0018/stdout.log
E086	.vulcan/engineering/run-0001/rtl-timing-lint.json
E087	.vulcan/engineering/run-0001/quartus-sta.json
E088	.vulcan/engineering/run-0001/board-hil.json

C Silicon signoff certification

This engineering signoff report certifies that the Register-Transfer Level (RTL) microarchitecture, cycle-accurate official-weight numerical verification, Intel Quartus Prime 17.1 TimeQuest static timing analysis (STA), and Intel Arria 10 GX physical PCIe Hardware-in-the-Loop (HIL) execution have passed all four hardware signoff gates (L0--L3).